



A Brief Introduction to Artificial Intelligence



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- 1. What's AI**
2. A Brief History of AI
3. AI Industry Ecosystem
4. Challenges to Operationalizing AI
5. AI Trends

What Is Your Understanding of AI?



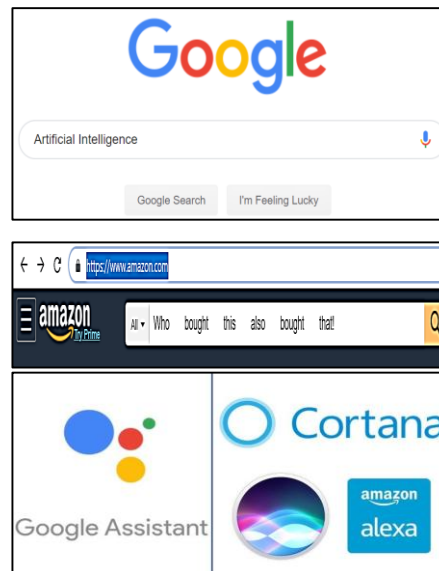
1

Terminator



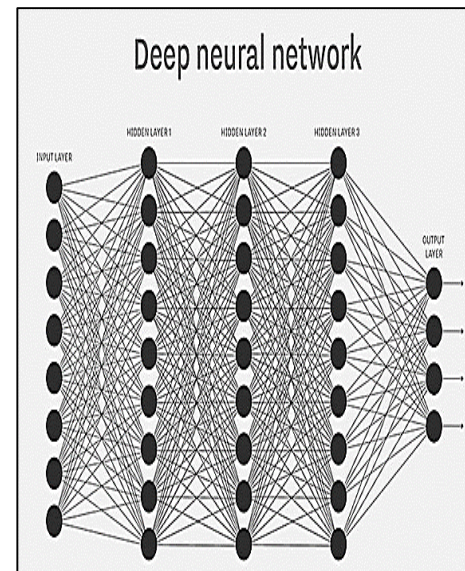
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Hyperscale
data center



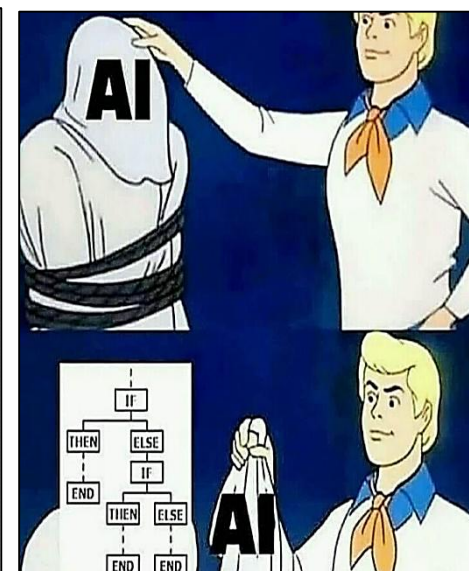
3

Intelligent
apps



4

Neural
network



5

if-else

What's AI?

Artificial Intelligence (AI) is the science (including theory, methods, techniques, and application systems) of mimicking, extending, and augmenting human intelligence through intelligent machines.

"...making a machine behave in ways that would be called intelligent if a human were so behaving."

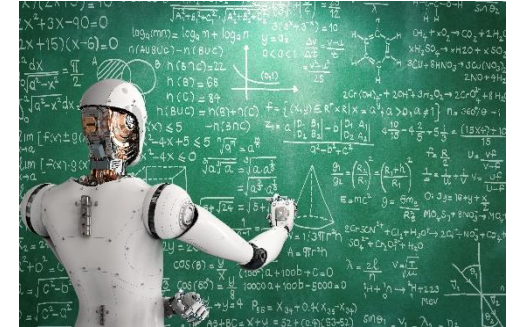
-- John McCarthy 1956

"Think humanly, think rationally, act humanly, and act rationally."

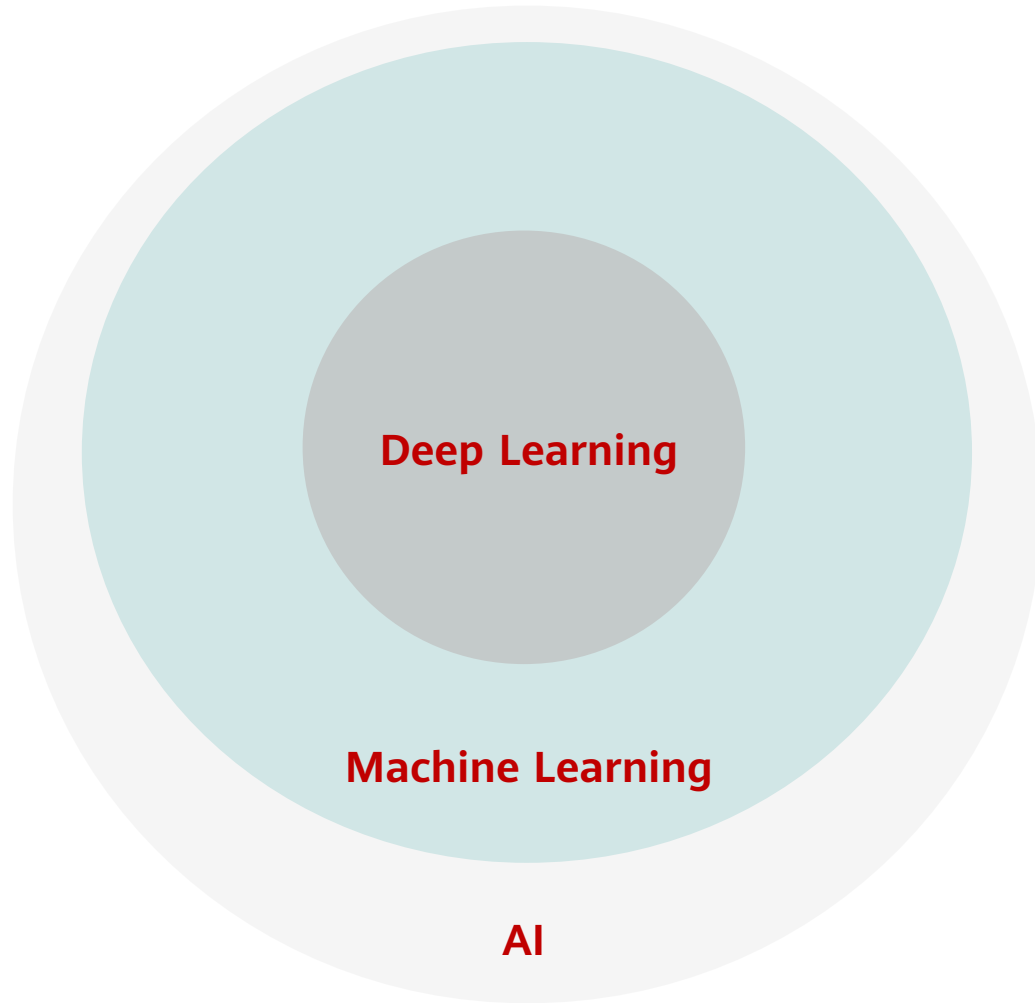
-- Stuart J. Russell

"AI is an intelligent entity created by humans or the branch of computer science that aims to create it."

-- Zhongdao Xiuzhi



AI vs. Machine Learning vs. Deep Learning



AI is the intelligence exhibited by an artificial entity.

Machine Learning (ML) is the core of AI and the fundamental way to make computers intelligent.

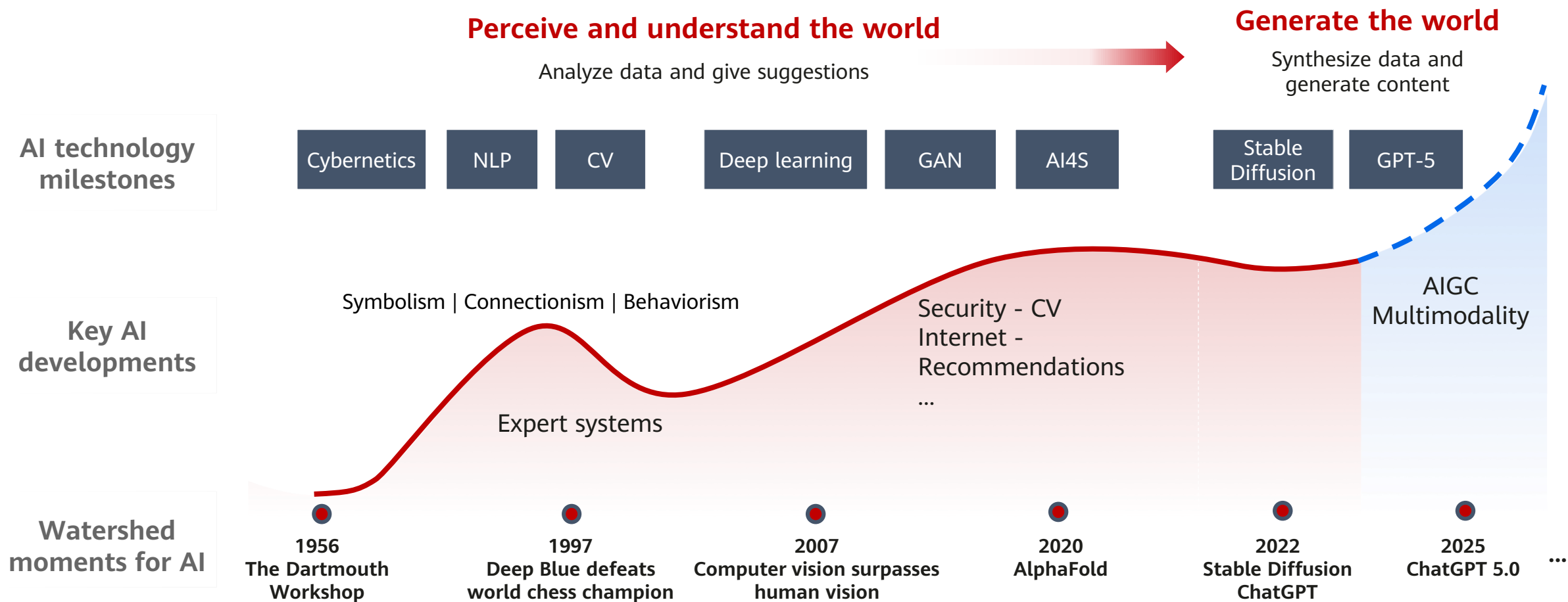
Deep Learning (DL) is a subset of machine learning. Neural networks in deep learning simulate a human brain by taking in large amounts of data and attempting to learn from it. DL has been a focus of AI in recent years.



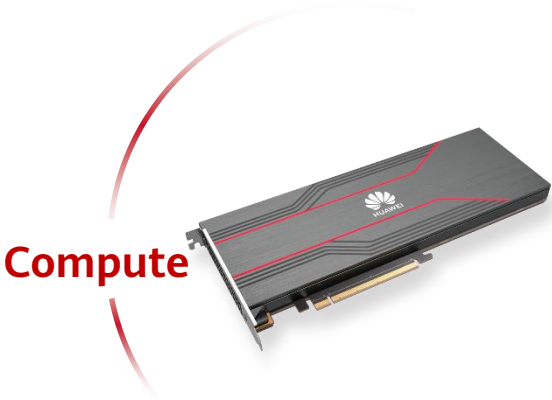
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A Brief History of AI



3 Factors Powering Deep Learning and the Explosion of AI



Ubiquitous computing power

Intelligent devices are penetrating life and industries.

1 GOPS
Intel Pentium III
computer

»»»

10 TOPS
Smartphone

Explosive data growth

Rich applications bring explosive growth of unstructured data.

44 ZB
2020

»»»

180 ZB
2025

Fast model iteration

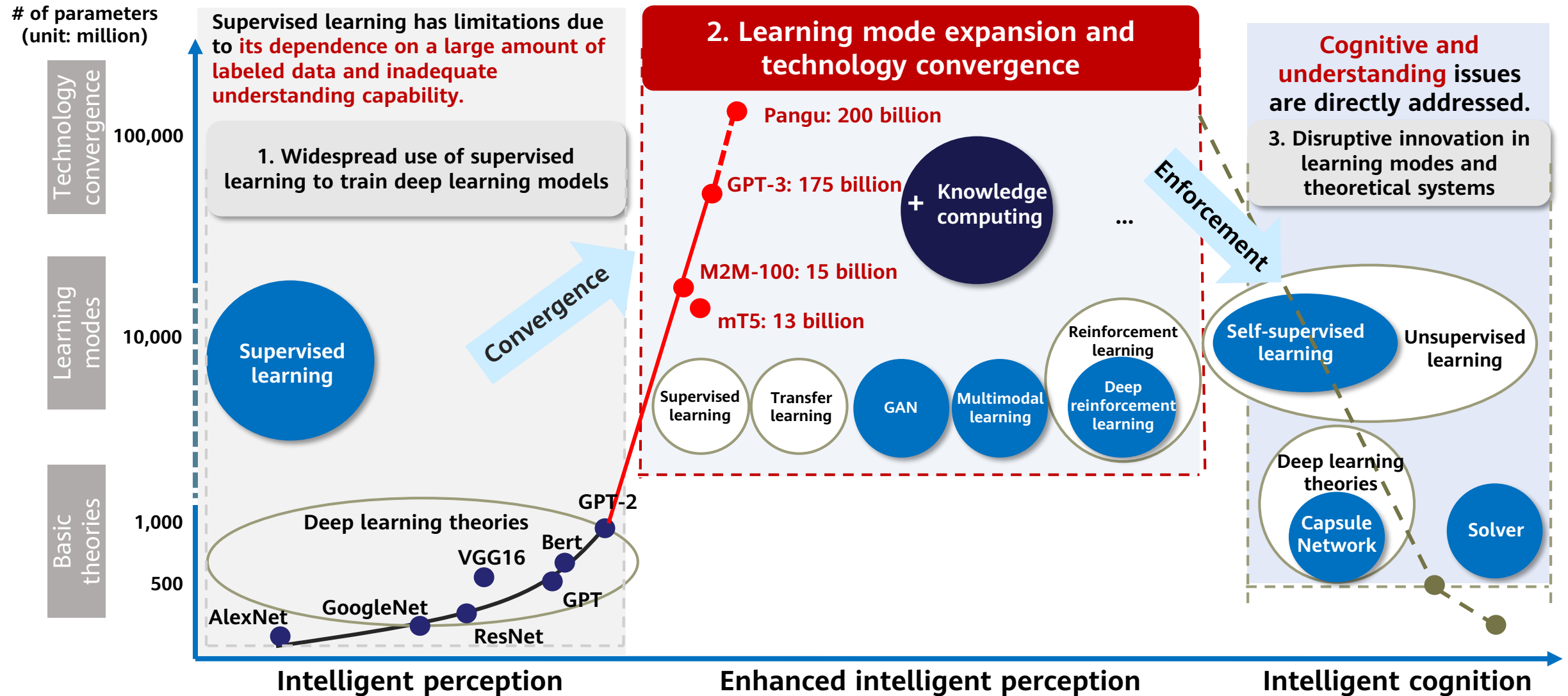
Models are getting bigger and bigger

110M
parameters for GPT-1

»»»

200B
parameters for the
PCL-L model

AI Technology: From Intelligent Perception to Intelligent Cognition



AI Technology: From Analytical AI to Generative AI

Today Analytical AI: Learning from data

- **Characteristics:** Analytical AI utilizes machine learning algorithms to **learn the underlying data distribution** and to classify, predict, group things and detect exceptions.
- **Typical technologies:** classical machine learning, fine-tuning, and neural networks such as convolutional neural network (CNN), deep residual network (DRN), and recurrent neural network (RNN)
- **Typical applications:**
 - ✓ Recommendation system: mining associations between users and items
 - ✓ Facial recognition: identifying a person based on facial feature information
 - ✓ Text recognition: recognizing text based on images containing text
 - ✓ Autonomous driving: traffic status monitoring and automatic parking



Tomorrow Generative AI: Learning from data + Generating data

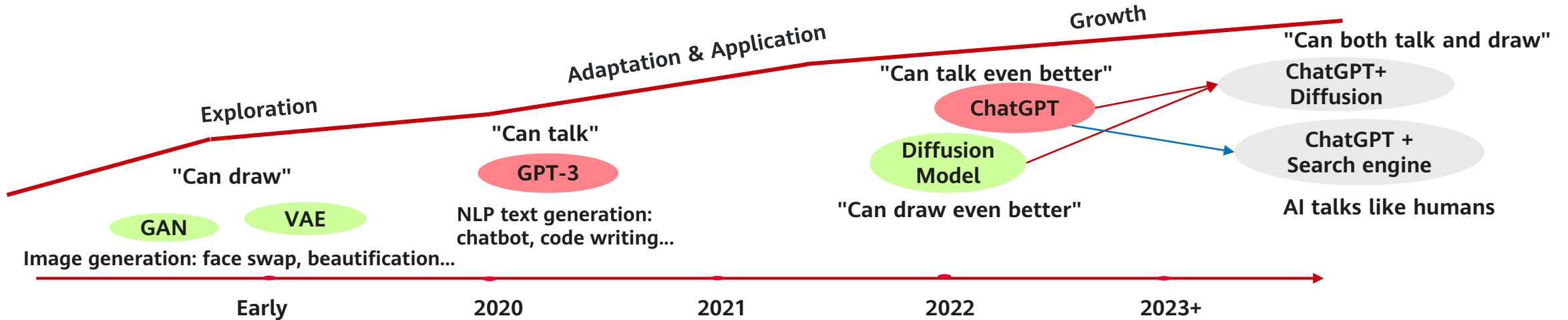
- **Characteristics:** Generative AI **generates new data samples** by **learning the underlying data distribution and understanding how data is generated**.
- **Typical technologies:** Transformer, Diffusion, and Prompting
- **Typical applications:**
 - ✓ Text creation: generating a complete copy based on prompt text
 - ✓ Code generation: generating complete code based on the context
 - ✓ Image generation: generating images in various styles based on key information, such as those used in blogs and posters
 - ✓ Multi-modal generation: text-to-image generation



AIGC: AI Generated Content

Development of AIGC

- Image generation: Typical image generation models are based on Diffusion models.
- Natural language dialog: ChatGPT is a typical representative model, which is Transformer-based.



According to QbitAI's forecast, **text-to-image/video AI models** are expected to have the largest market share by **2030**. **China's AIGC market size is expected to exceed CNY1 trillion by 2030.**

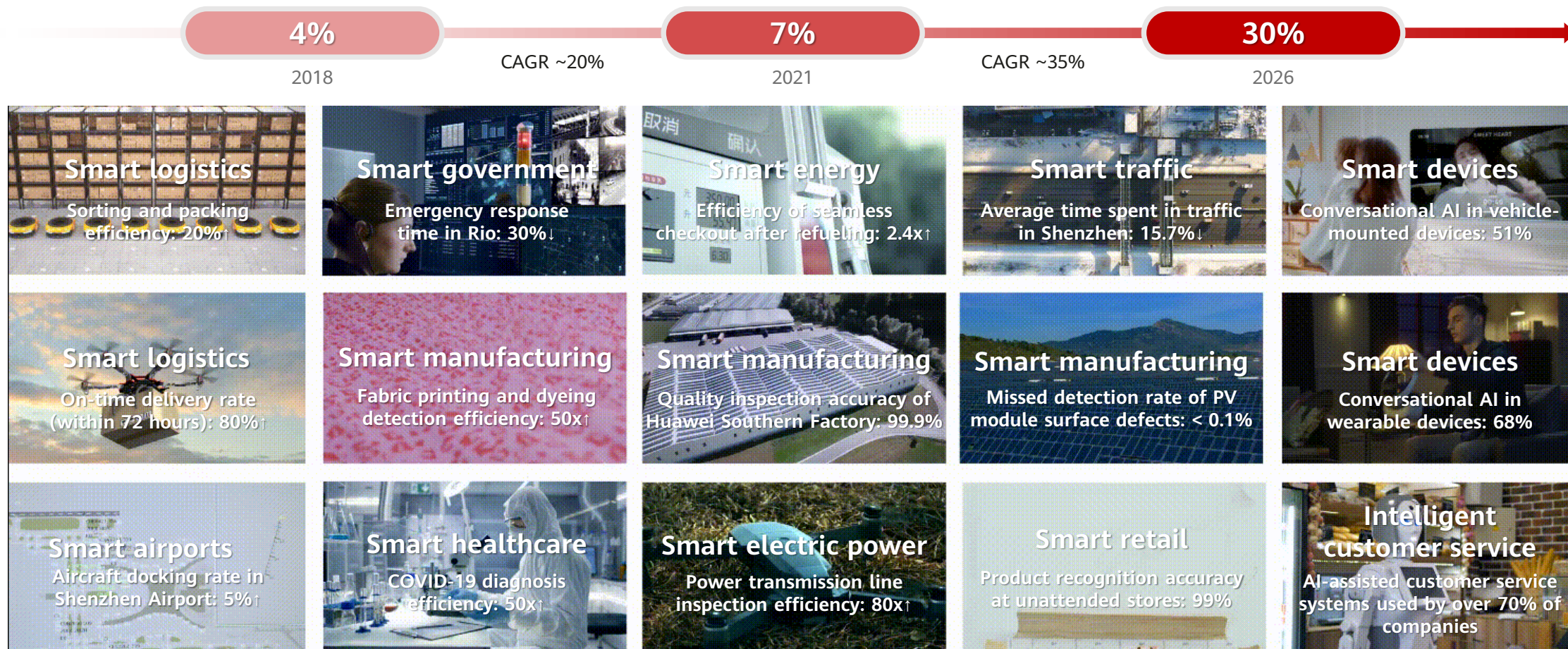


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AI Adoption Is Skyrocketing

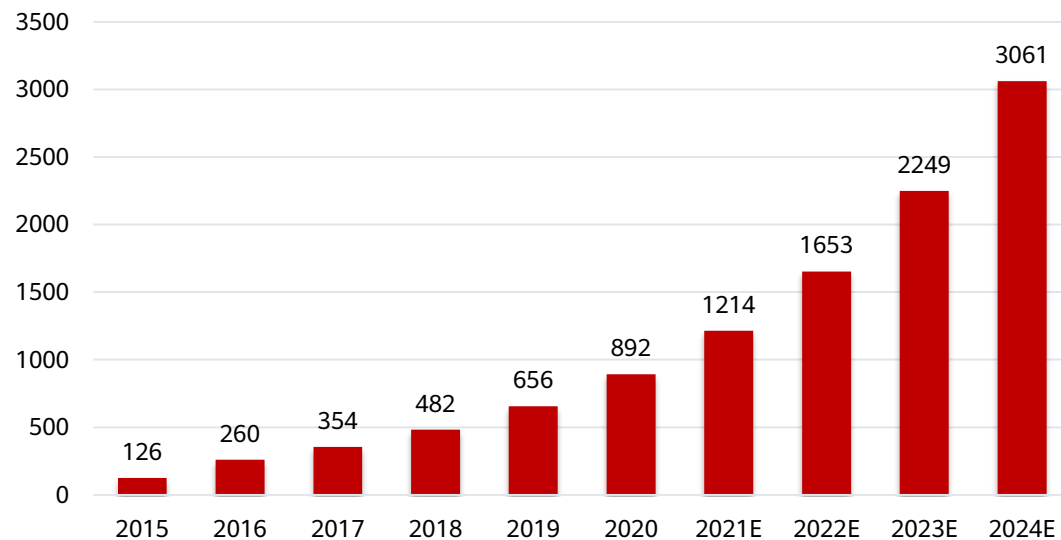
AI adoption is expected to reach 30%+ by 2026.



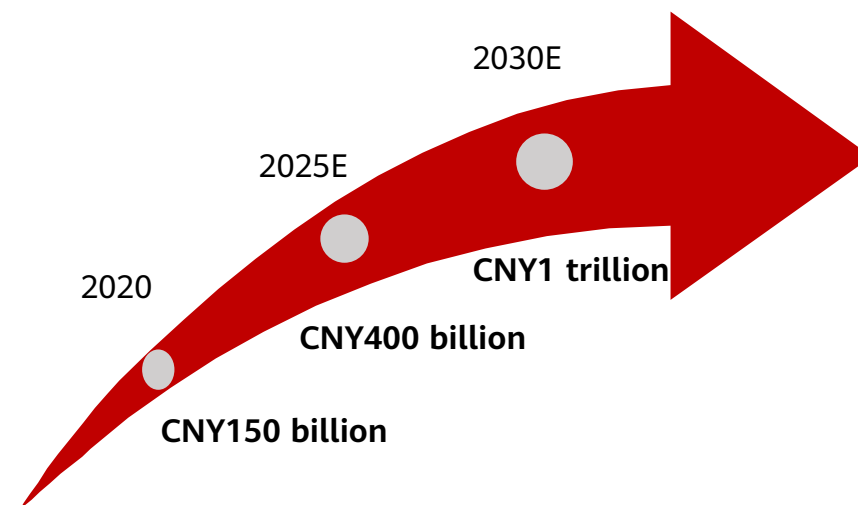
Predictions about the AI Software Market

The global AI market is growing rapidly. The market in 2022 was estimated at **USD165.3 billion**. China's AI market is also seeing steady growth.

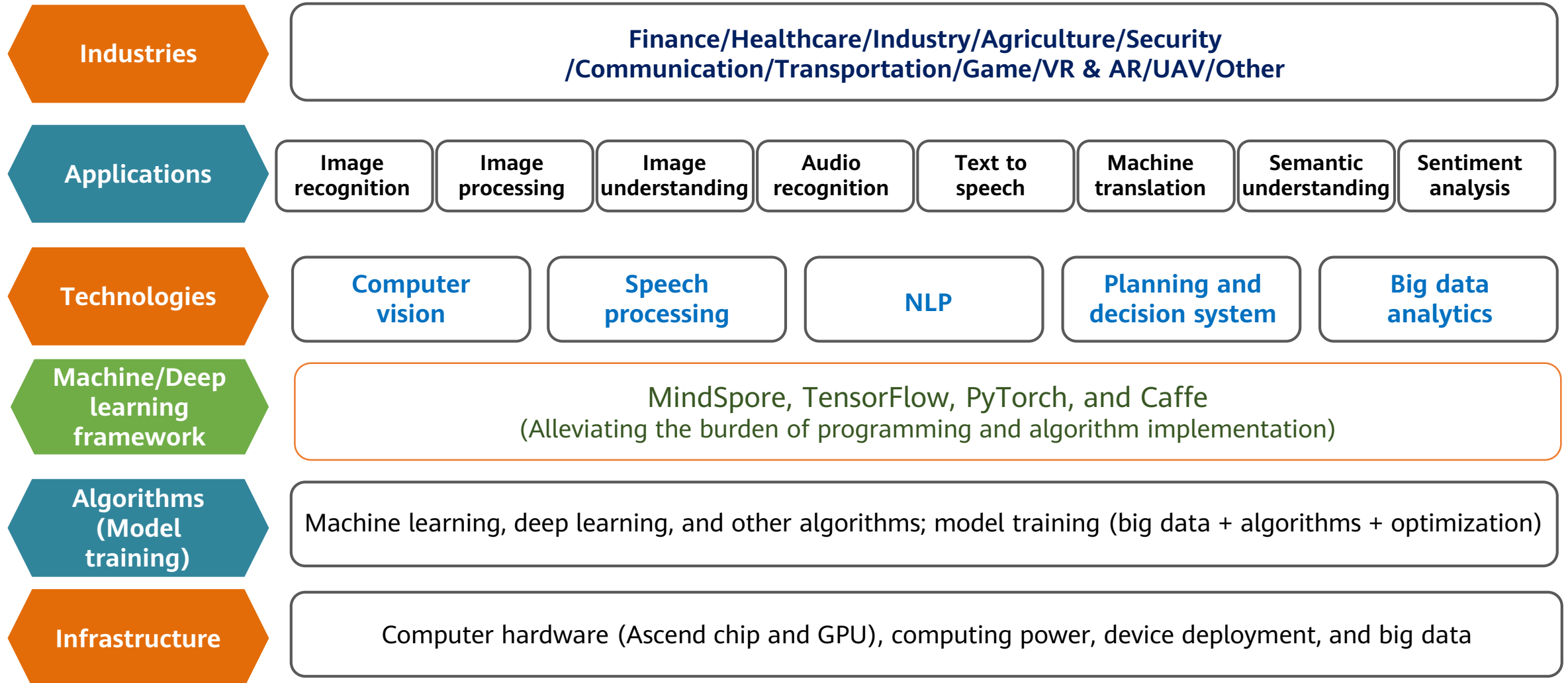
**World's AI market from 2015 to 2024
(billion USD)**



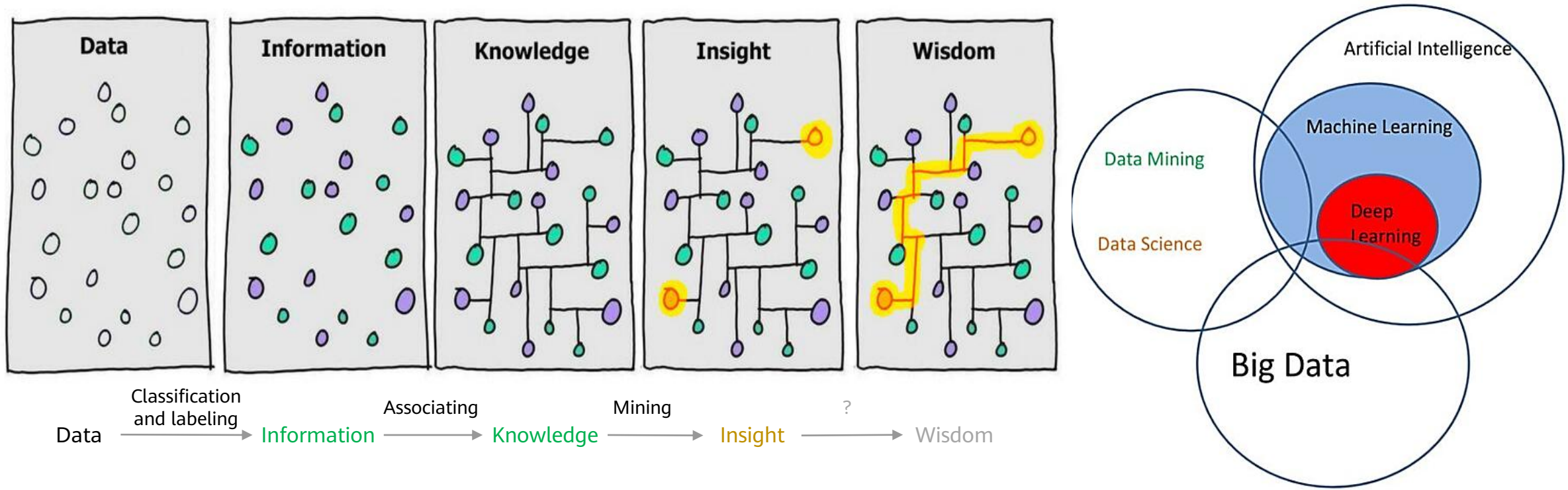
**China's AI market from 2020 to 2030
(CNY)**



AI Technology Architecture

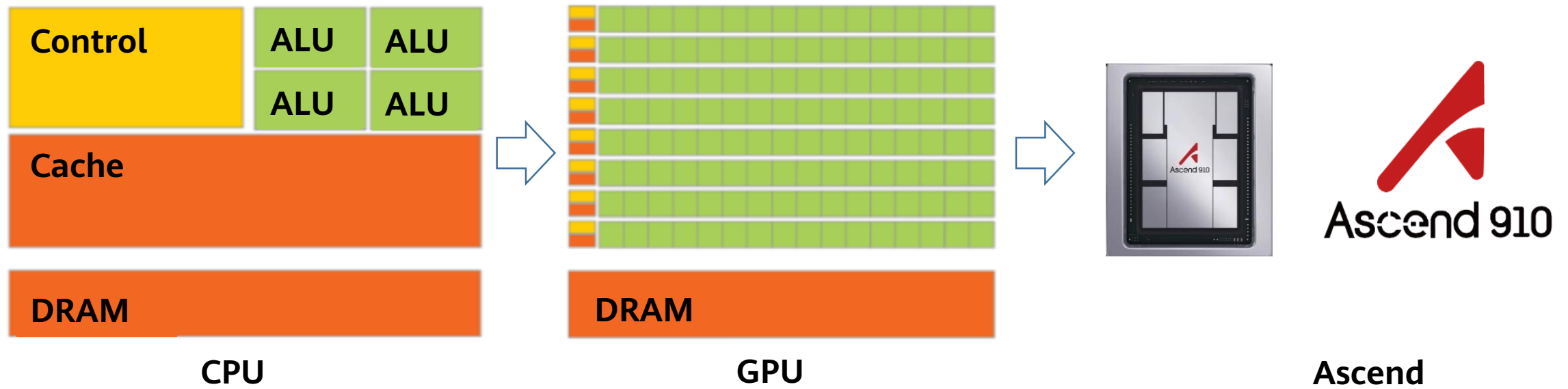


Infrastructure — Big Data



Data is an abstract expression of different phenomena in the real world. An enormous amount of data (big data) is generated every second worldwide from various sources such as netizens, mobile device users, machines, servers, labs, sensors, and devices. We can process this vast amount of big data and mine valuable information, knowledge, and even wisdom from it. Various AI services can utilize the acquired knowledge.

Infrastructure — Computer Hardware



We've seen a shift from CPUs and GPUs to dedicated AI chips, such as Ascend AI chips, to provide significantly increased computing power. While GPUs are more versatile, dedicated AI chips have a simpler architecture and more advanced capabilities.

Algorithms — Machine Learning

Input: 0, 8, 15, 22, 38

Output: 32, 46.4, 59, 71.6, ?

Question 1: What is the output corresponding to input 38?

Question 2: Write a beauty pageant program.

```
if (height == a &  
    weight == b &  
    face_shape == symmetric &  
    eye_diameter >= c &  
    nasal_height >= d &  
    body_shape == ...  
    ...  
)  
    return beauty //Rule-based beauty pageant  
algorithm
```

The process you're using to find the answer in your mind is similar to a **machine learning process**, where you're trying to find rules from input-output pairs.



Algorithm

def function(C):

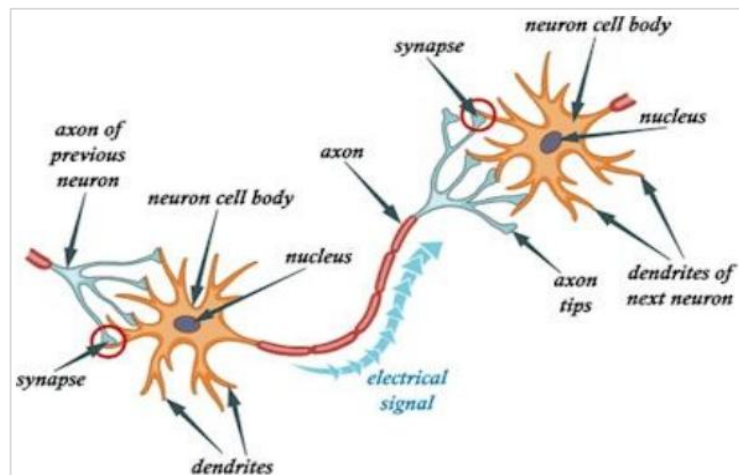
$F = C * 1.8 + 32$

return F

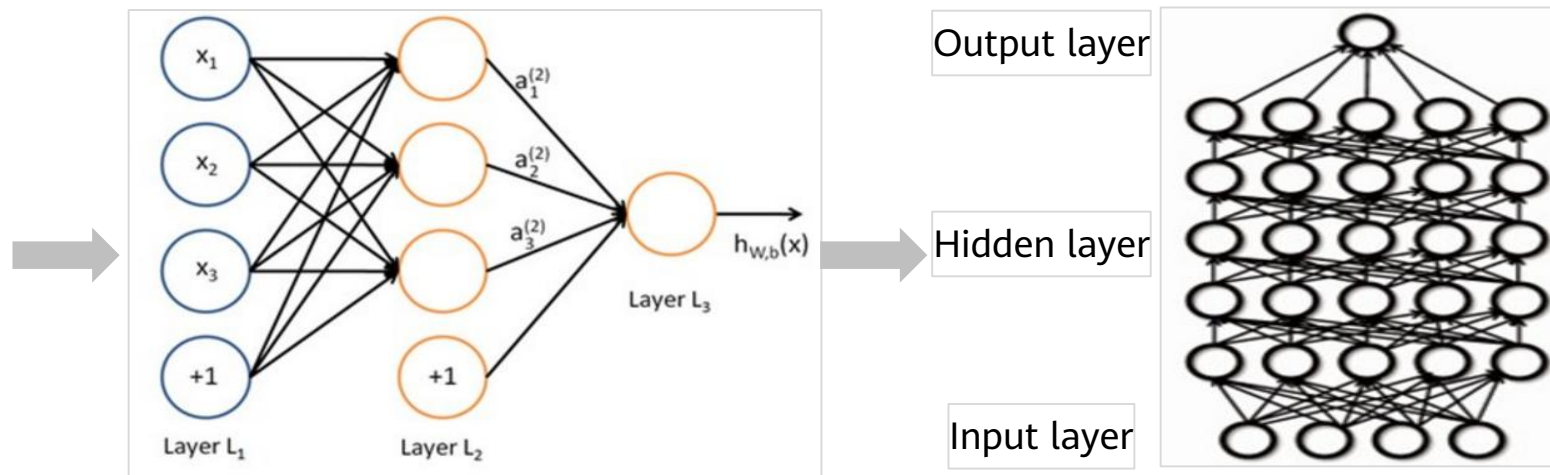
Input

Output

Algorithms — Neural Network and Deep Learning

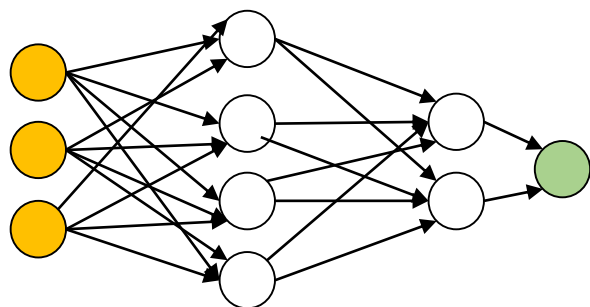


Human neural network

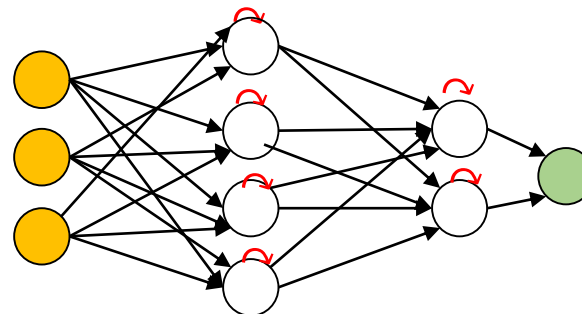


Perceptron

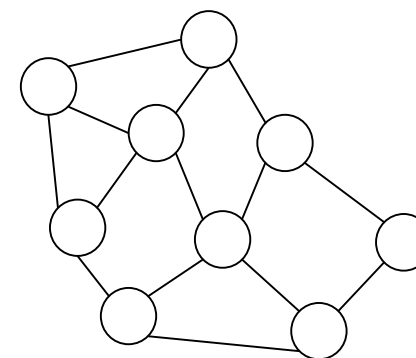
Deep neural network



(a) Feedforward neural network (FNN)



(b) Feedback network



(c) Graph network

Technologies — Computer Vision



Content moderation

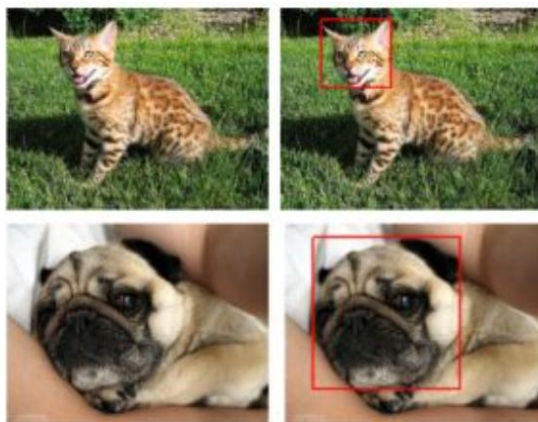


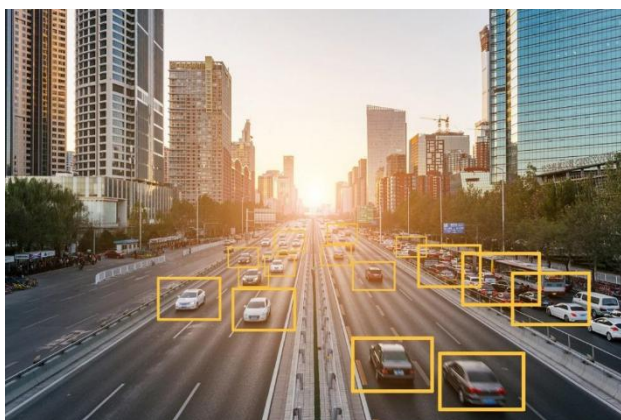
Image search



Semantic segmentation



Face detection

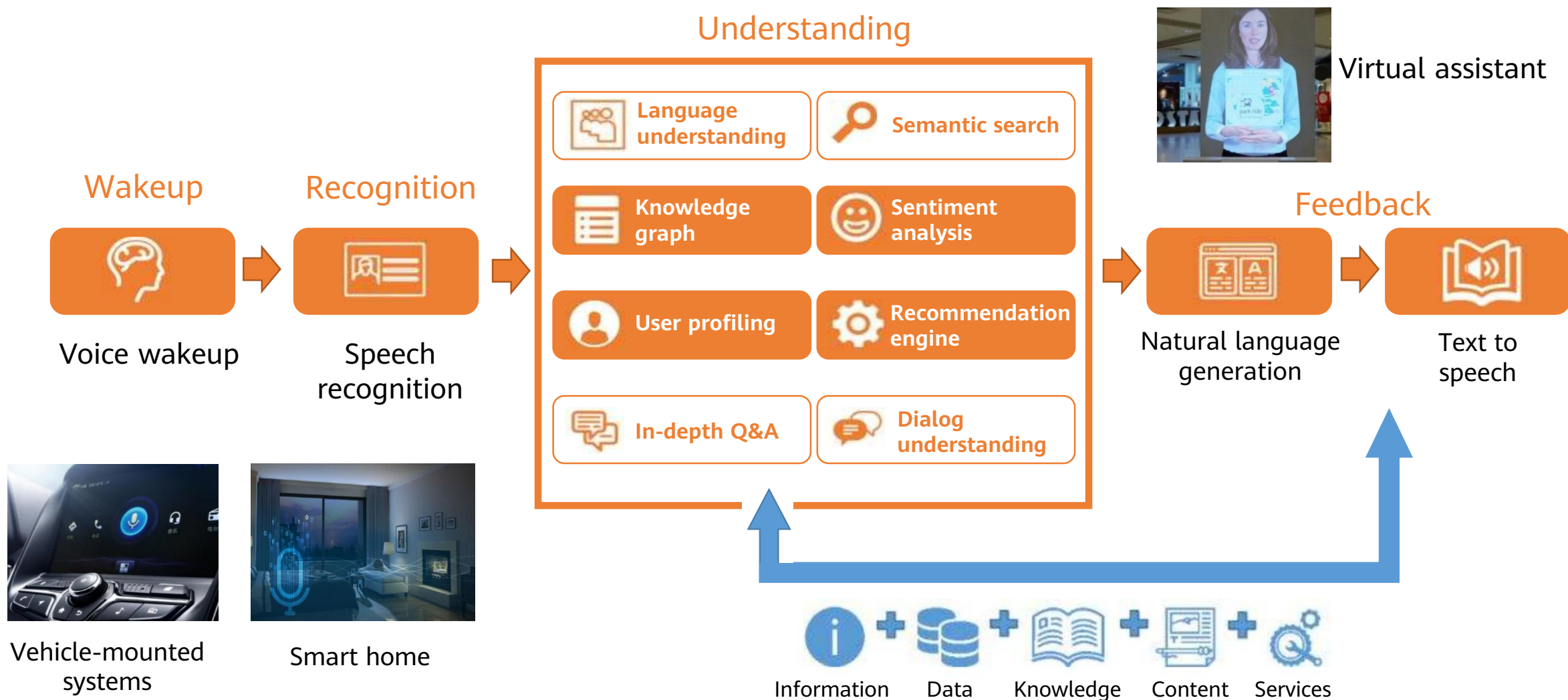


Object detection

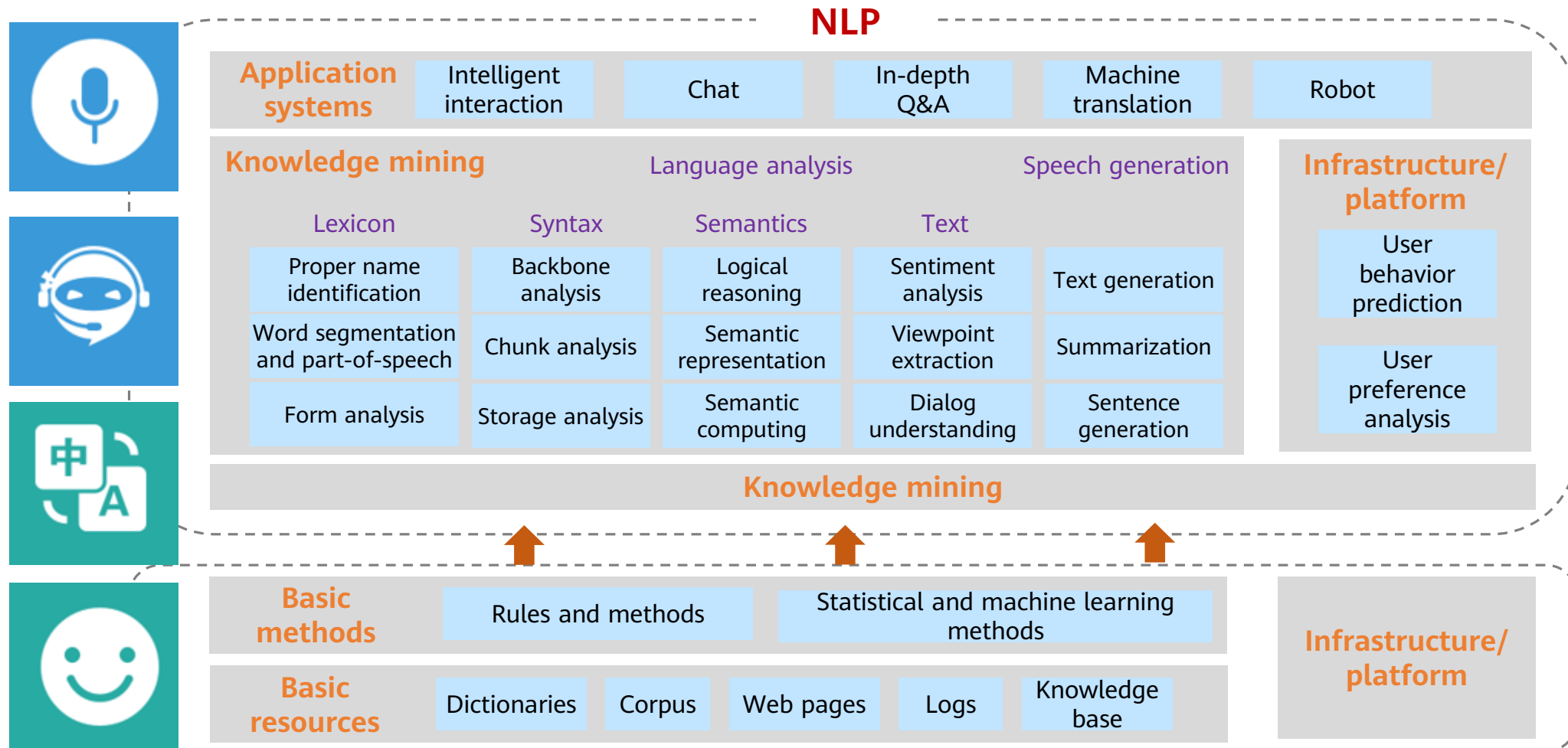


Image classification

Technologies — Speech Processing



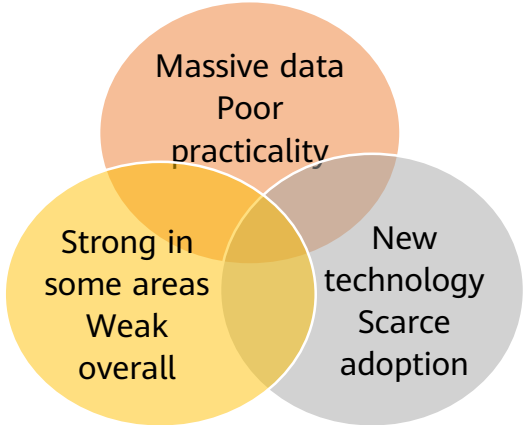
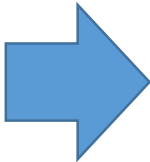
Technologies — Natural Language Processing



Applications — AI in Smart Cities



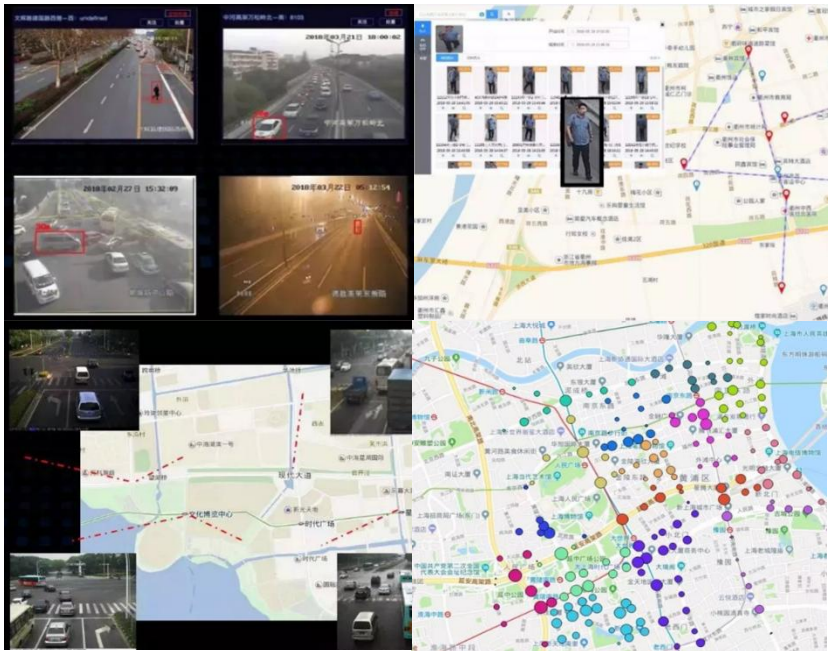
Excessive hardware but insufficient data



City-wide perception

City-wide warning

City-wide traffic dashboard



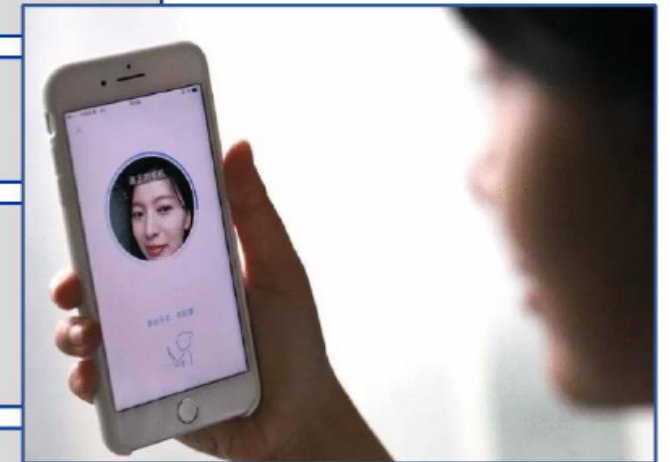
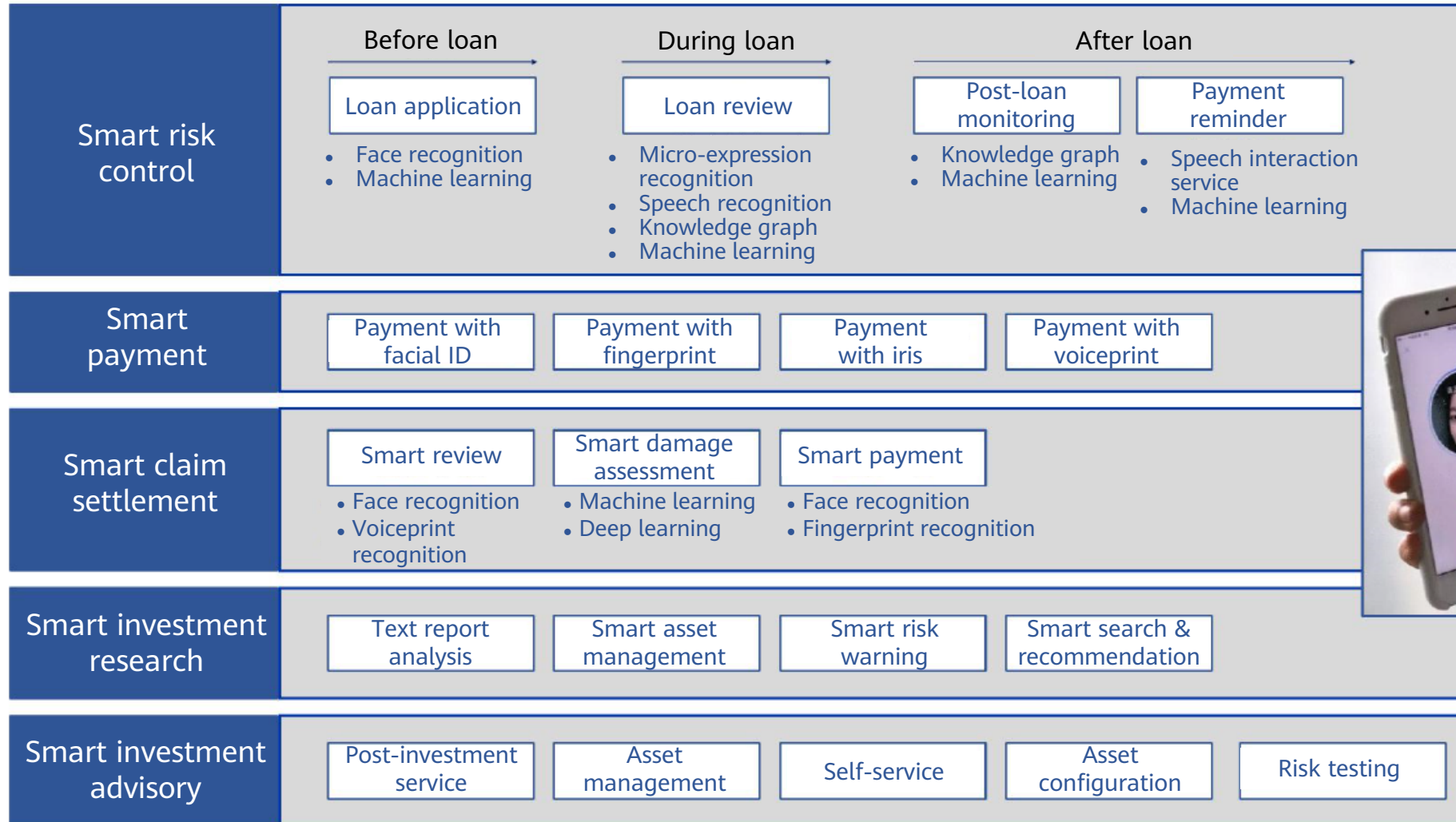
City-wide search

City-wide collaboration



Emergency vehicle monitoring

Applications — AI in Finance



Applications — AI in Retail



Design

- Display color recommendation
- Intelligent image analysis



Manufacturing

- Sample photographing
- Intelligent quality check



Profit management

- Optimal pricing strategy



Supply chain

- Warehousing design
- Inventory management
- Zoning
- Transport route optimization



E-commerce

- Search by image
- Intelligent recommendation



Brick & mortar store retail

- Intelligent site selection
- Intelligent goods recognition
- Intelligent monitoring
- Optimized shelf display
- Customer flow statistics
- Consumer behavior analysis

From the Internet

Applications — AI in Medicare



Case recording via voice

- Efficiently records doctor-patient communication.
- Drives the digitalization of medicare.

Medical imaging analysis

- Automatically identifies and marks lesions.
- Helps doctors easily detect rare symptoms.

Comprehensive treatment

- Serves as a virtual doctor.
- Uses NLP, knowledge graphs, and CV to facilitate diagnosis and treatment.

Physical health management

- Monitors and manages health status, and predicts the occurrence of diseases.

Medical robots

- Examples: surgical robots and rehabilitation robots.
- Improves surgical precision.

Drug R&D

- Accelerates the discovery of candidate drugs.
- The technology, once matures, is expected to cut the cost of new drug R&D by US\$28 billion annually.

Applications — AI in Agriculture



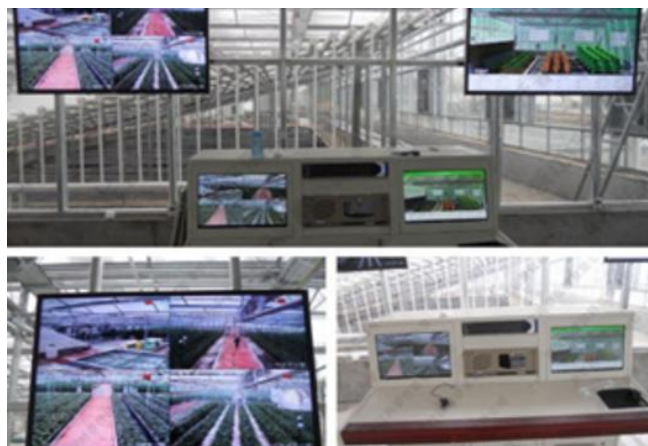
Intelligent agriculture data analysis and predictions



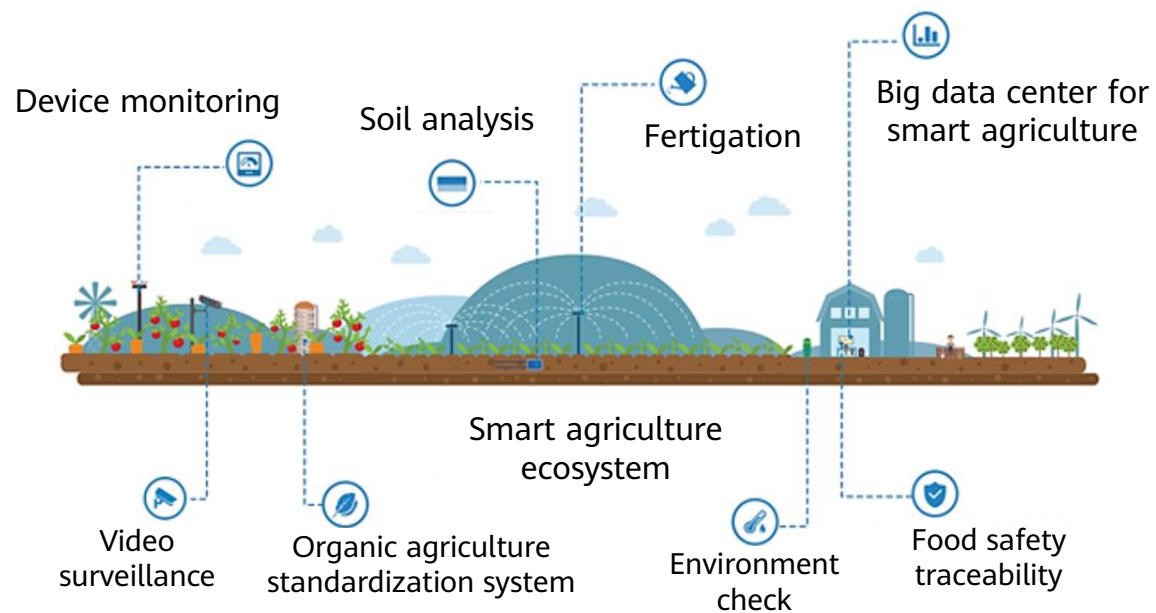
Pesticide cost reduction



Harvesting cost reduction



Airborne control system





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Challenges in Terms of Training Data

Data acquisition

Insufficient data
Poor data quality
Data security & compliance
Data ownership

Data governance

Multi-source heterogeneous data
Unstructured data
Massive data storage and applications

Data labeling

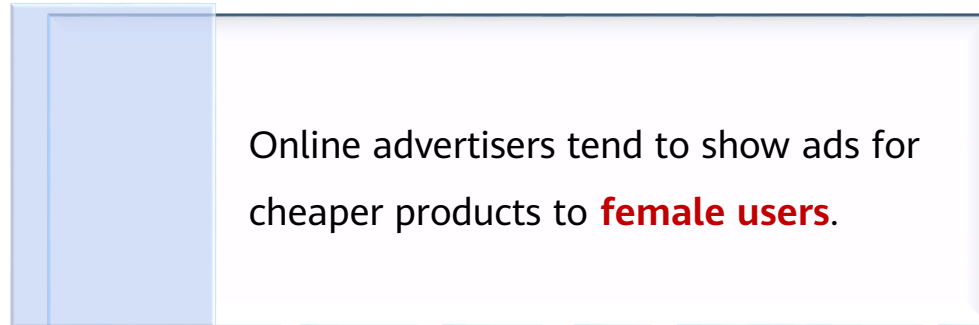
Data collection for rare cases
Complex scenario understanding
Data security

Poor Explainability

- Today, AI is being used in many mission-critical systems in our society, but we still haven't figured out how AI works, especially how foundation models work.
- The poor explainability of existing deep learning systems compromises people's confidence in the security and stability of current AI systems. An important topic of AI research for today and the future is to study how to use AI for the good of humanity and prevent any harmful results.

AI Bias

- **Any bias in AI comes from the data it is trained on.**
- AI bias refers to a phenomenon where AI systems produce biased results that reflect and perpetuate human biases within a society, including historical and current social inequality. Examples include biased decisions that reflect discriminatory tendencies, such as on the basis of race, gender, or other factors. AI can even make biased decisions simply based on a person's name or address.



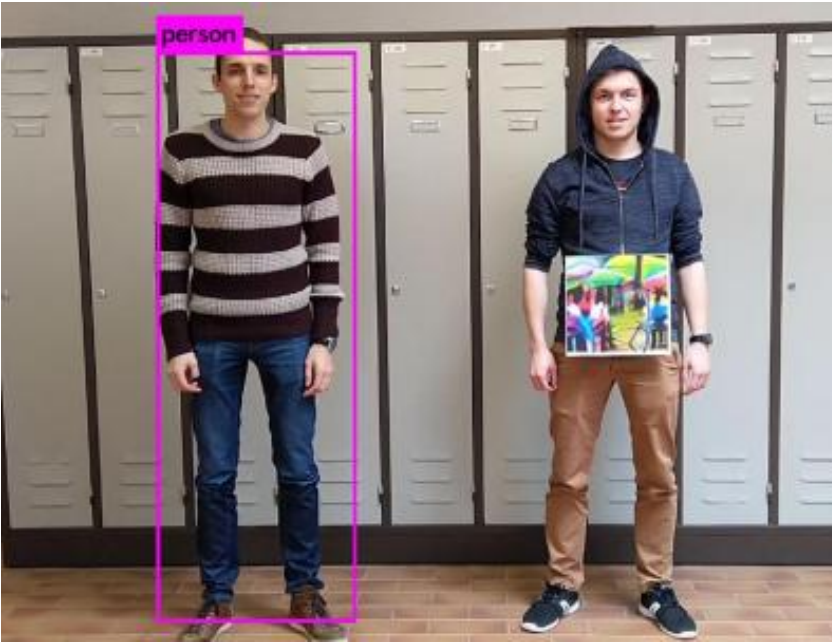
Privacy Issues

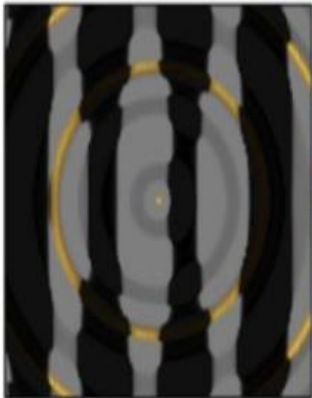
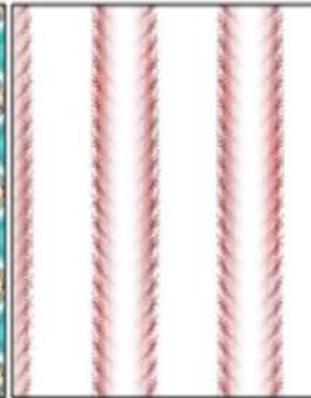
- Existing AI algorithms are all data-driven, as they rely on massive amounts of training data. While we enjoy the convenience enabled by AI and other digital technologies, all our data is being collected by technology companies whose apps we're using, such as Facebook, Google, Amazon, and Alibaba. These data tell a lot about us, including our political views, religion, and personal lives.

When we browse the Internet, technology companies can record everything that we do: every click or tap, every page scrolling, the time, and the browsing history.

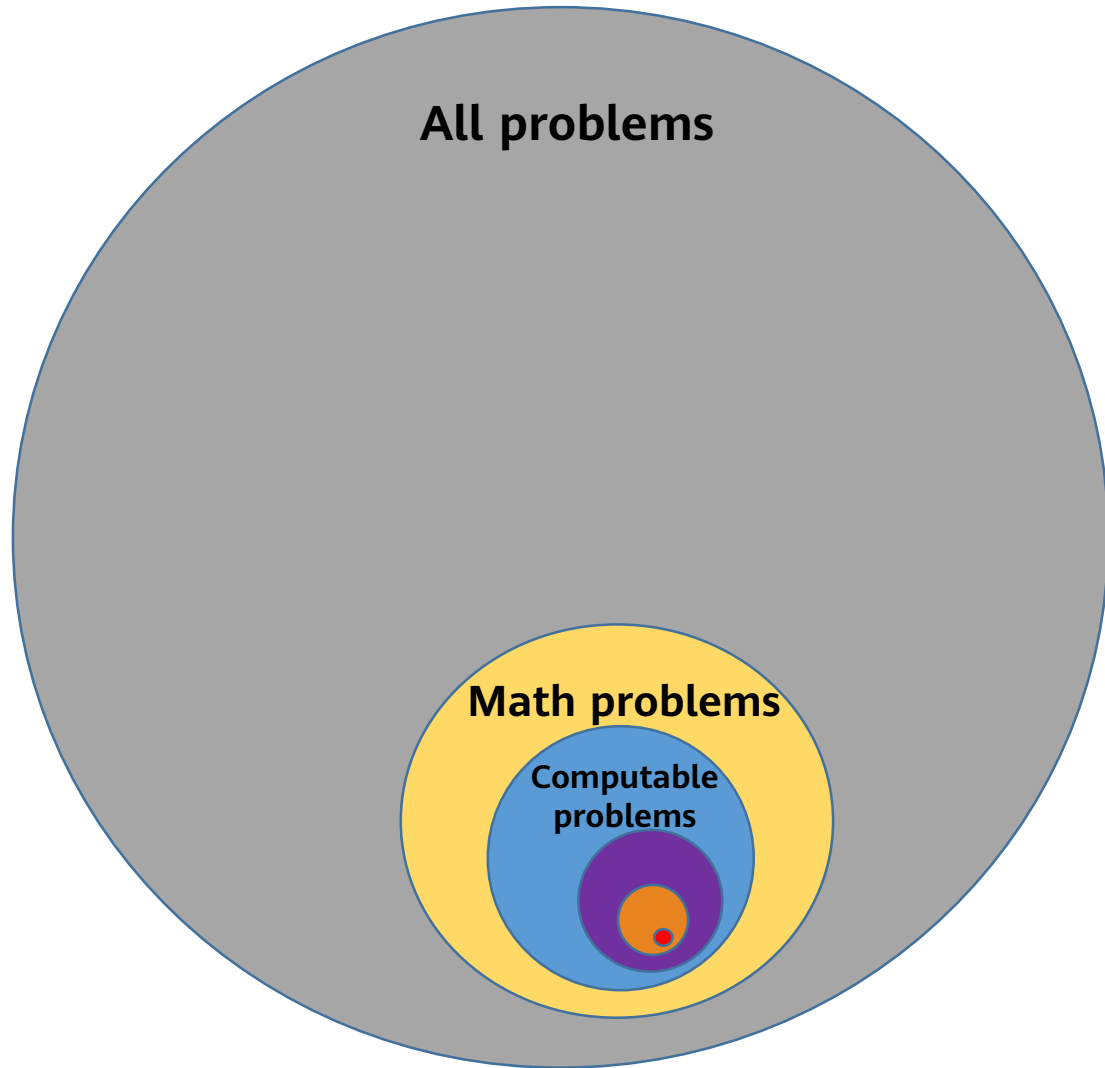
A ride hailing platform can know more about us than we normally expect: where we went, where we are, what we did, our education, consuming power, personal preferences, and more.

Examples of AI Making Mistakes



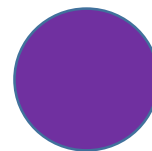
<i>BELIEVE WITH $\geq 99.6\%$ CERTAINTY</i>			
			
centipede	peacock	jackfruit	bubble
			
king penguin	starfish	baseball	electric guitar

Will AI Surpass Human Intelligence in the Future?



Turing computability theory:

1. Are there definitive answers to all math problems in the world?
2. If there are, can the answers be calculated using a limited number of steps?
3. For a math problem that can be solved in a limited number of steps, is there some kind of machine that can keep working on that problem until it finally solves it?



Problems solvable to a Turing machine



Problems solvable to AI



Problems already solved by AI

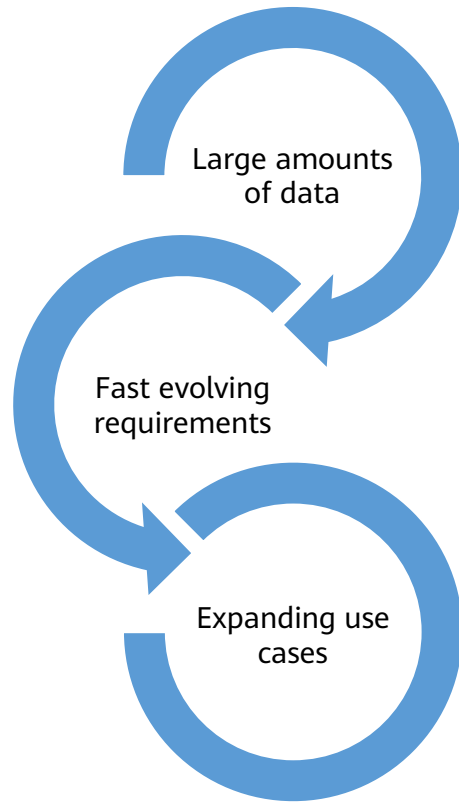


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More Complex AI Basic Data Services

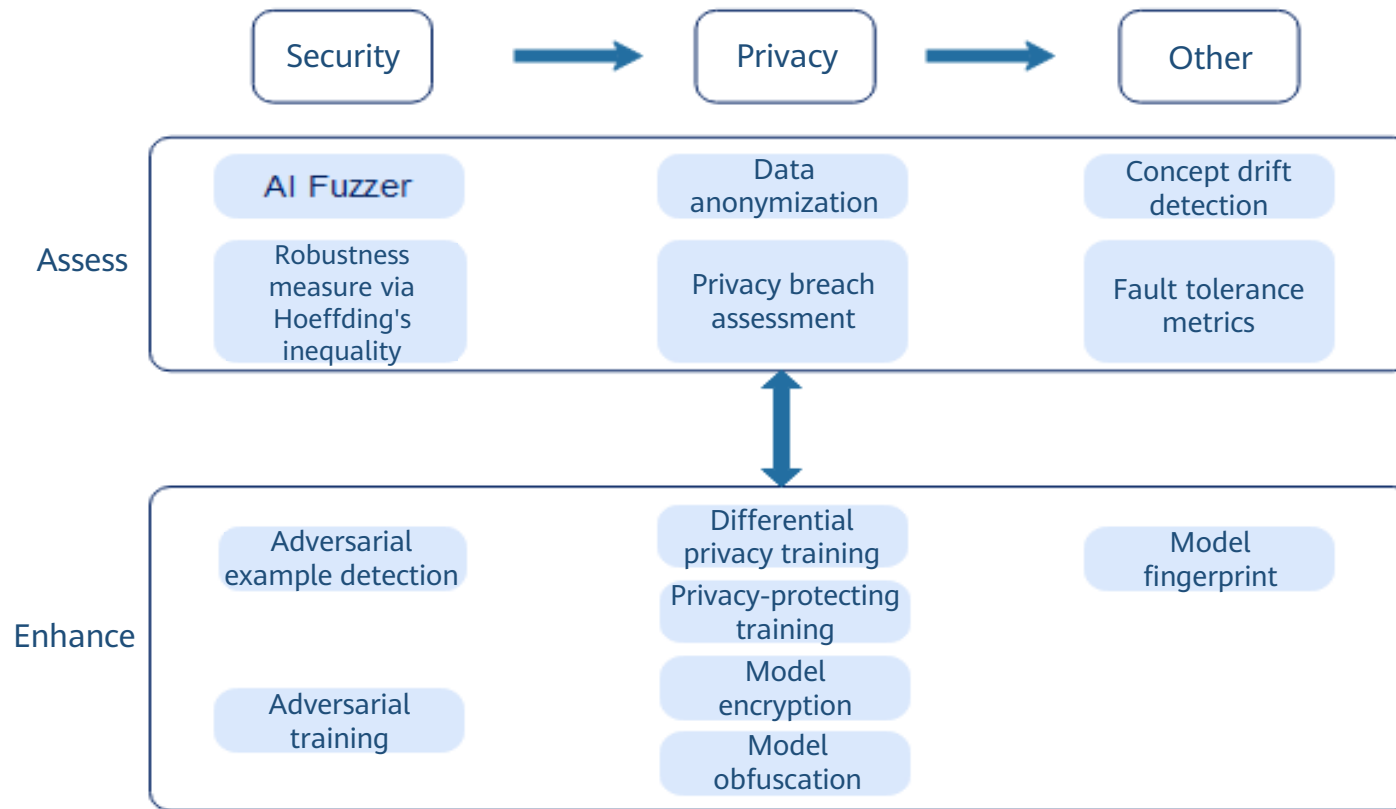
- Trends in AI basic data services: complex, automated, full-stack, and compliant



- Manual data labeling is costly. AI-automated data labeling is increasingly used by basic data service providers and AI algorithm companies for its lower costs and higher efficiency.
- As AI algorithms are iterating and evolving rapidly, the requirements for data labeling, including labels and the dimensions of information, are also evolving, making data labeling more challenging for AI data service providers.

More Secure Data Sharing

- Federated learning is a way of training AI models without the need to pool data into a centralized location. It protects data security while allowing developers to obtain more data to train their models.



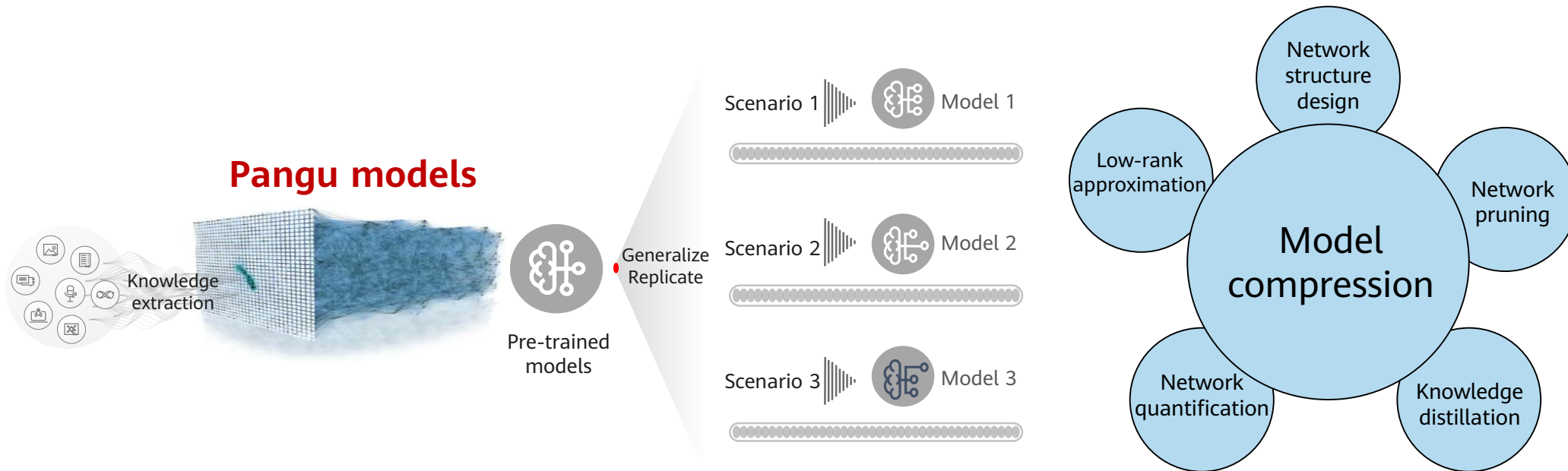
Collaborative Computing Power from End Devices, the Edge, and the Cloud

- AI chips are becoming more powerful each day, whether on the cloud, the edge, or end devices.



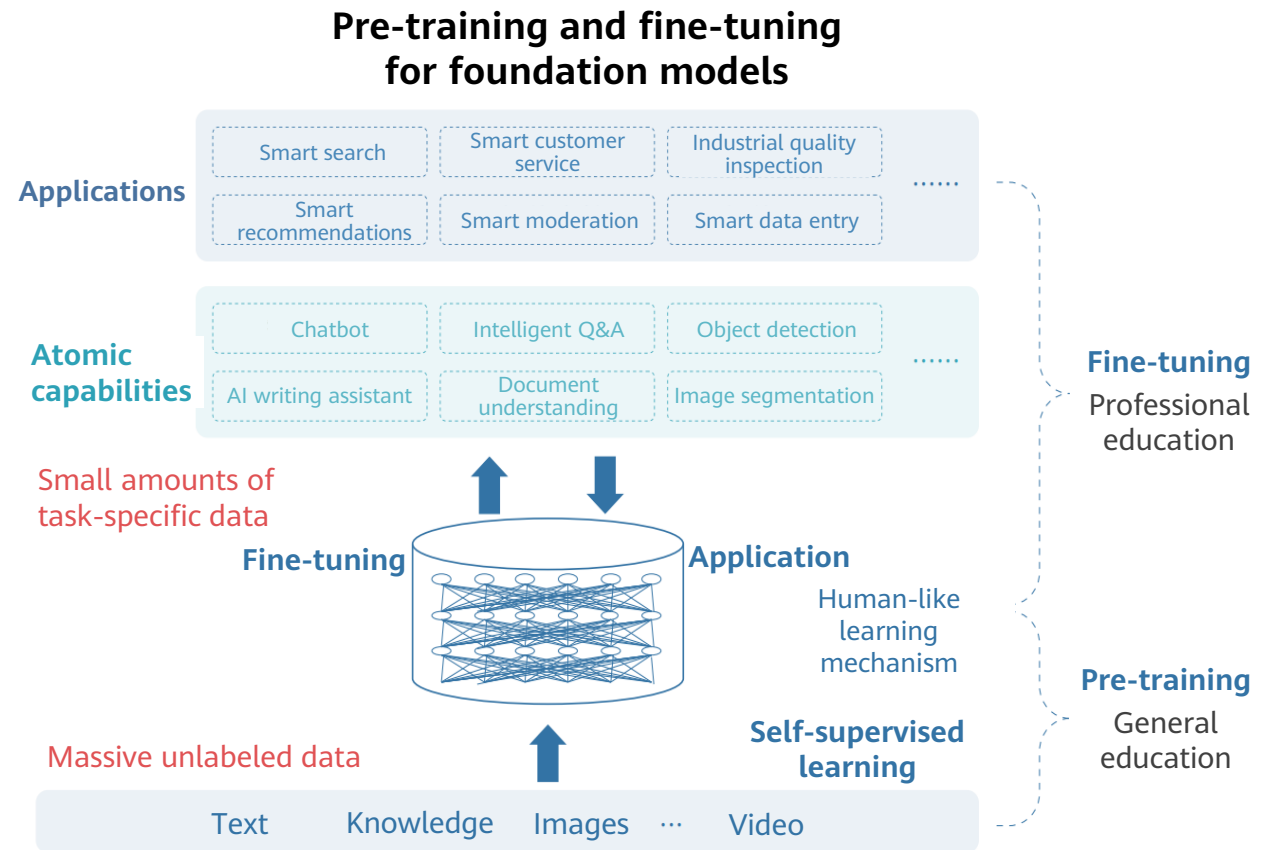
Larger Models and Distillation

- On one hand, AI models, powered by big data and fast growing computing power, are becoming bigger and bigger, with more parameters, delivering better generalization performance. On the other hand, to be able to deploy on the edge or mobile devices, smaller models must be extracted from large models using a technique called model distillation.

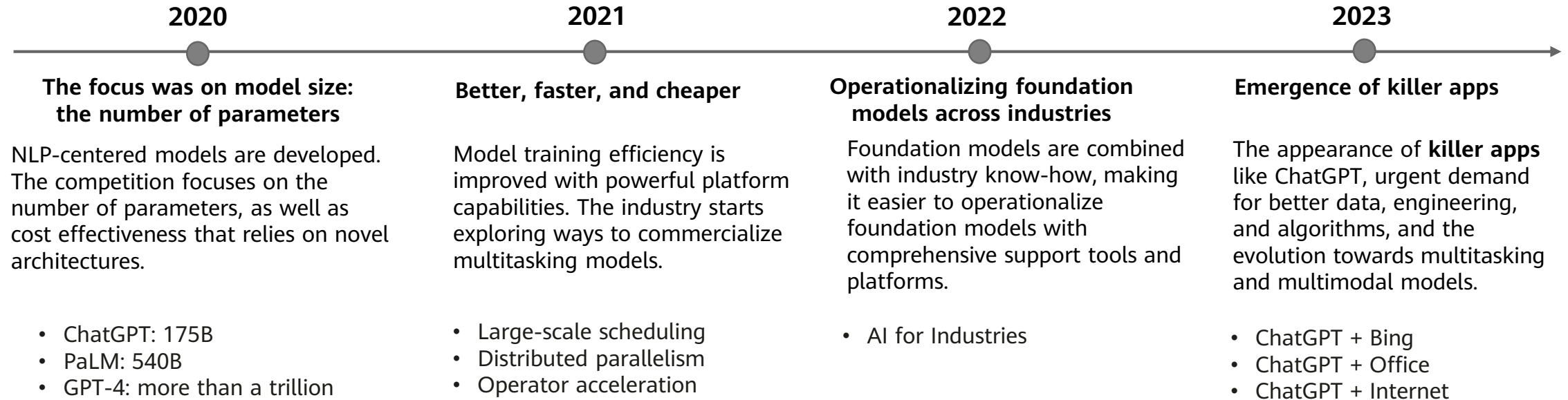


What Are Foundation Models?

- A foundation model is typically a large-scale deep learning model with billions or even trillions of parameters. Currently, the term most commonly refers to large language models (LLMs).
- Foundation models are large-scale AI models pre-trained on broad, diverse datasets. They represent a significant upgrade from previous, task-specific algorithms. Through open-source frameworks or open APIs, developers can quickly adapt such models to downstream tasks through zero-shot or few-shot learning, achieving superior performance in recognition, understanding, decision-making, and content generation, while enabling lower-cost AI development and deployment.



Future Trends in Foundation Models

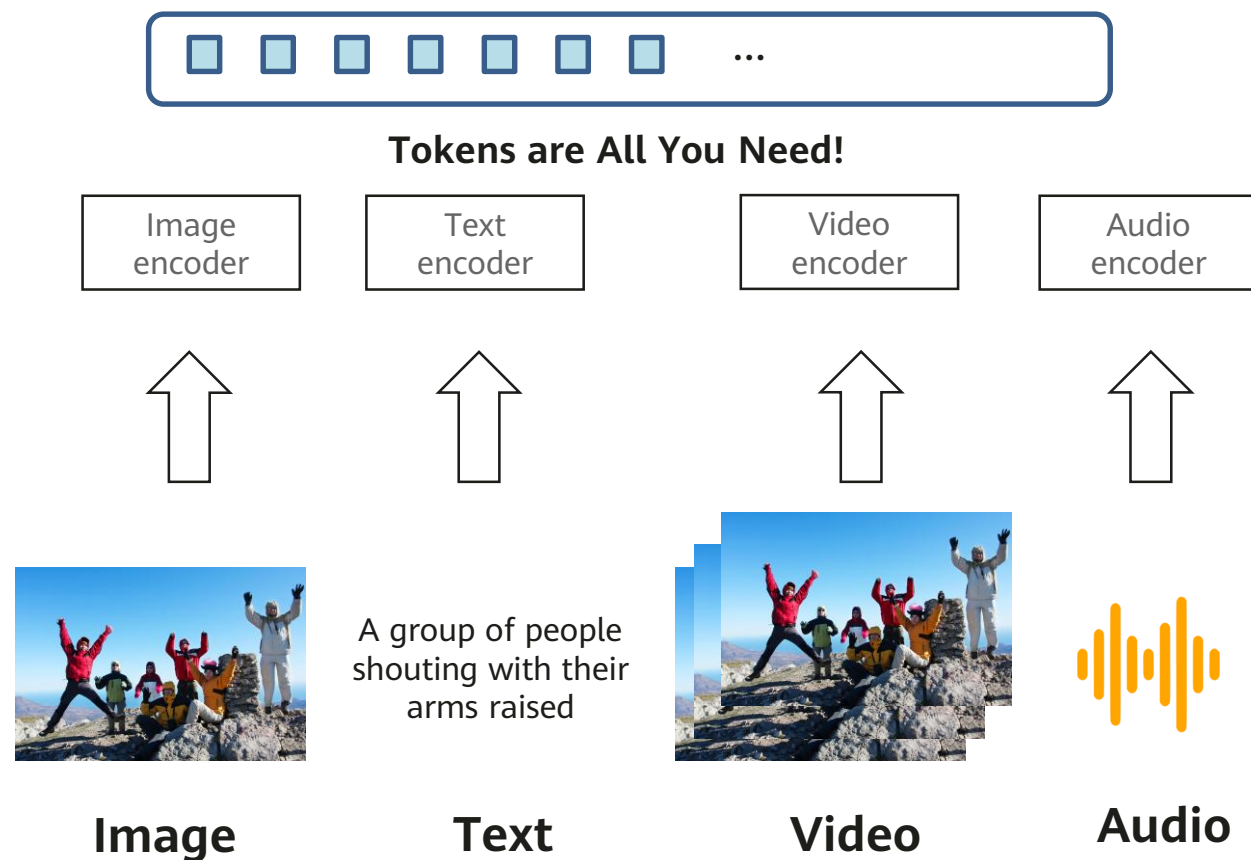


Trends

Models are still getting larger	Models tailored to industry needs	Multimodal convergence
Some models have reached 500 billion parameters and are still growing linearly. They are approaching or even surpassing the number of neuron connections in the human brain. Examples: OpenAI GPT-3 (175B) and Google PaLM (540B)	Industry-tailored models (L1) can be obtained by further training L0 models (general foundation models) on industry datasets. L1 models are better able to address specific industry needs. Example: coal mine model, electric power model, and drug molecule model. SenseTime: INTERN system (beginner-expert-generalist); Baidu: industry models	In the next 2-3 years, CV and NLP models are expected to converge into multimodal models with a unified architecture. There is also deep convergence between different foundation models. Examples: text+visual, voice+text, and text+video. OpenAI DALL·E/E2, Google Imagen, and Microsoft GLIP.

Unified Multimodal Perception Models for Multiple Types of Tasks

- Natural language is a highly abstracted semantic space shaped by human cognition.
- Using natural language as the central interface, we can map diverse features from mainstream modalities into a unified semantic space.



Easier-to-Use Frameworks

- AI development frameworks are becoming easier to use and more versatile, continuously simplifying AI development. Examples of such frameworks include MindSpore, TensorFlow 2.x, and PyTorch.



Fast Expanding Use Cases

AI is expected to be used to solve more problems in more areas. New AI use cases are fast emerging. They include:

- Psychotherapy
- Car damage assessment (insurance)
- Office automation
- ...



Replika: My AI Friend

Luka, Inc Health & Fitness

★★★★★ 1,15,078

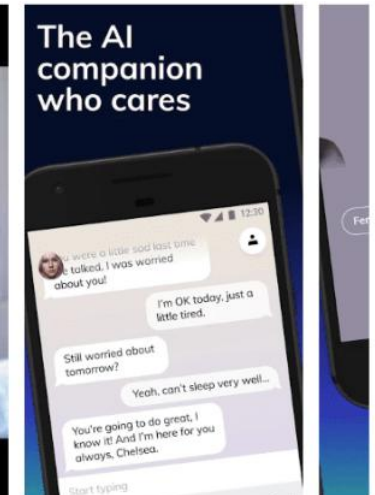
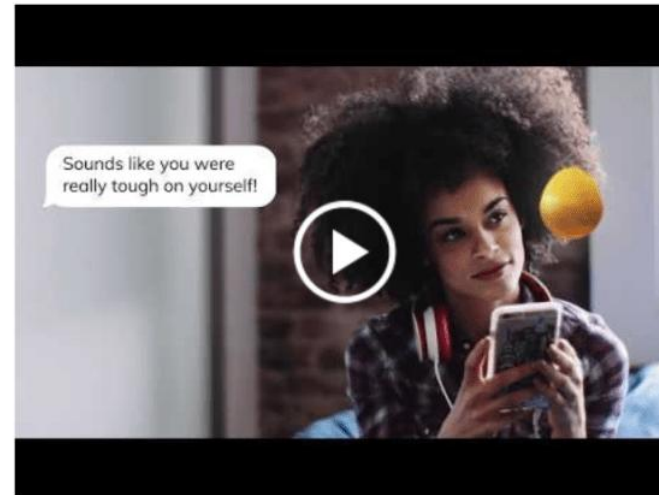
Mature 17+

Offers in-app purchases

This app is compatible with your device.

Add to wishlist

Install





Section Summary

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Q&A

1. (Multiple-answer question) Which of the following are the four elements of AI? ()
 - A. Data
 - B. Algorithms
 - C. Computing power
 - D. Scenarios
 - E. Frameworks



2. (Multiple-answer question) Which of the following are different phases of AI? ()
- A. Intelligent computing
 - B. Intelligent perception
 - C. Intelligent cognition
 - D. Intelligent sensing



Acronyms and Abbreviations

- CPU: central processing unit
- GPU: graphic processing unit
- ISV: independent software vendor
- CV: computer vision
- NLP: natural language processing
- CNN: convolutional neural network
- RNN: recurrent neural network



Thank You.

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